

Differential effect of oleic and palmitic acid on lipid accumulation and apoptosis in cultured hepatocytes.



BACKGROUND AND AIM: Studies have shown monounsaturated oleic acid to be less toxic than palmitic acid and to prevent/attenuate palmitic acid hepatocytes toxicity in steatosis models in vitro. However, to what degree these effects are mediated by steatosis extent is unknown. METHODS: We evaluated whether steatosis per se is associated with hepatocytes apoptosis and determined the role of oleic and palmitic acid, the most abundant fatty acids in western diets, on triglyceride accumulation and apoptosis in an in vitro model of steatosis induced in three hepatocytic cell lines (HepG2, HuH7, WRL68). The impact of incubation for 24 h with oleic (0.66 and 1.32 mM) and palmitic acid (0.33 and 0.66 mM), alone or combined (molar ratio 2 : 1) on steatosis, apoptosis, and insulin signalling, was evaluated. RESULTS: Concurrent with PPARGamma and SREBP-1 gene activation, steatosis extent was larger when cells were treated with oleic than with palmitic acid; the latter fatty acid was associated with increased PPARalpha expression. Cell apoptosis was inversely proportional to steatosis deposition. Moreover, palmitic, but not oleic acid, impaired insulin signalling. Despite the higher amount of fat resulting from incubation of the two fatty acids combined, the apoptosis rate and impaired insulin signalling were lower than in cells treated with palmitic acid alone, indicating a protective effect of oleic acid. CONCLUSIONS: Oleic acid is more steatogenic but less apoptotic than palmitic acid in hepatocytic cell cultures. These data may provide a biological basis for clinical findings on dietary patterns and pathogenetic models of nonalcoholic fatty liver disease.